



PRESIDENCY UNIVERSITY

BENGALURU

EVEN SEMESTER – (2025-2026), SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

ANNOUNCEMENT OF ENDTERM EXAMINATION

Course Title	: Machine Vision for Robotics	Course code	: ECE3107
Instructor In-charge	: Dr. K. Gowri	Semester	: VI
Date	: 25-05-2026 (Monday)	Max. Marks	: 50
Timing	: 1:00PM – 4:00PM	Weightage	: 100%

SYLLABUS:

MODULE 1: Overview of Machine Vision in IP:

Basic Components – Elements of visual perception, Lenses: Pinhole cameras, Gaussian Optics – Cameras – Camera-Computer interface- Fundamental Data Structures: Images, Regions, Sub-pixel Precise Contours

Image Enhancement: Gray value transformations, image smoothing, Fourier Transform – Geometric Transformation - Image segmentation – Segmentation of contours, lines, circles and ellipses – Camera calibration

Stereo Reconstruction- Object recognition, Approaches to Object Recognition, Recognition by combination of views – objects with sharp edges, using two views only, using a single view, use of depth values.

MODULE 2: Vision algorithms and applications:

Transforming sensor reading, Mapping Sonar Data, aligning laser scan measurements - Vision and Tracking: Following the road, Iconic image processing, Multiscale image processing Video Tracking - Learning landmarks: Landmark spatiograms, K-means Clustering, EM Clustering.

MODULE 3: ROBOT Vision:

Basic introduction to Robotic operating System (ROS) - Real and Simulated Robots - Introduction to OpenCV, Open NI and PCL, installing and testing ROS camera Drivers, ROS to OpenCV - The cv_bridge Package.

IMPORTANT NOTE:

1. The conduction of examination (End-term) will be in offline mode in the University Campus.

Template:

10 Questions X 2 Marks = 10 Marks

4 Questions X 20 Marks (with subdivisions) = 80 Marks

Total Marks = 100 Marks

Signature of Instructor in charge